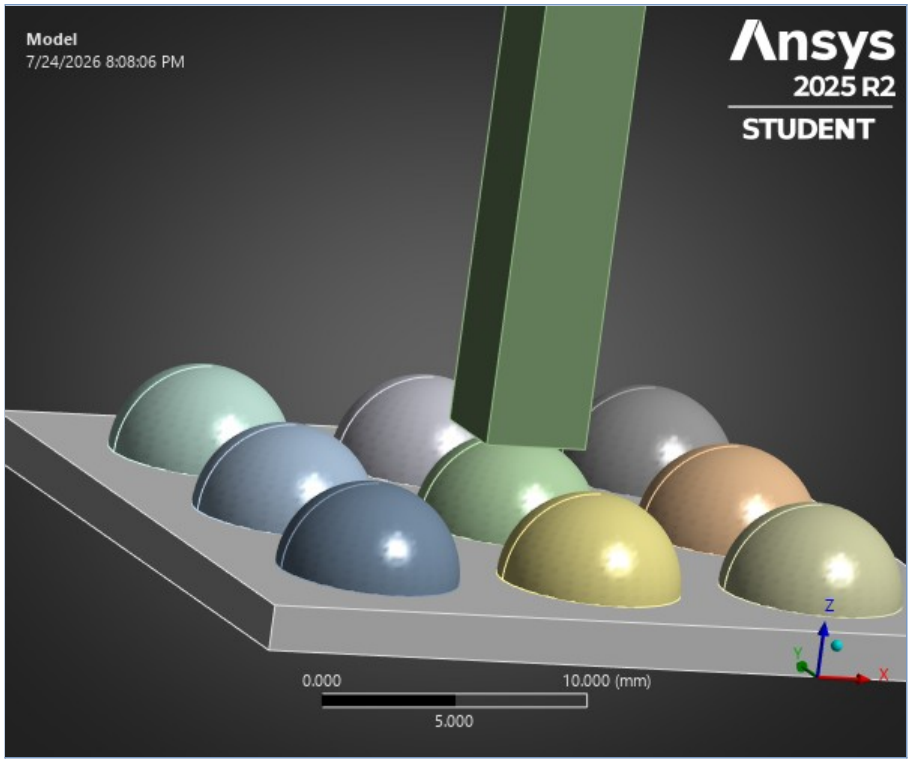
Project*

First Saved	Thursday, July 23, 2026
Last Saved	Thursday, July 23, 2026
Product Version	2025 R2
Save Project Before Solution	No
Save Project After Solution	No



Contents

- **Units**
- **Model (B4)**
  - Geometry Imports
    - Geometry Import (B3)
  - Geometry
    - Parts
  - Materials
  - Coordinate Systems
  - Connections
    - Contacts
      - Contact Regions
      - Frictionless - RUN 5|Body3 To RUN 5|Body11
  - Mesh
  - **Static Structural (B5)**
    - Analysis Settings
    - Loads
    - Solution (B6)
      - Solution Information
      - Results
      - Contact Tool
        - Status
      - Probes
- **Material Data**
  - ECOFLEX-0045
  - Plastic, ABS (high-impact)

Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (B4)

TABLE 2
Model (B4) > Geometry Imports

Object Name	*Geometry Imports*
State	Solved

TABLE 3
Model (B4) > Geometry Imports > Geometry Import (B3)

Object Name	*Geometry Import (B3)*
State	Solved
Definition	
Source	C:\Users\DLSU\Downloads\ANSYS FOR HOLLOW RODS\RUN 5.step
Type	Step
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No

Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (B4) > Geometry

Object Name	*Geometry*
State	Fully Defined
Definition	
Source	C:\Users\DLSU\Downloads\ANSYS FOR HOLLOW RODS\RUN 5.step
Type	Step
Length Unit	Millimeters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	30. mm
Length Y	30. mm
Length Z	43.3 mm
Properties	
Volume	2699.9 mm³
Mass	2.8437e-003 kg
Scale Factor Value	1.
Statistics	
Bodies	11
Active Bodies	11
Nodes	6869
Elements	2574
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (B4) > Geometry > Parts

Object Name	*RUN 5|Body1*	*RUN 5|Body2*	*RUN 5|Body3*	*RUN 5|Body4*	*RUN 5|Body5*	*RUN 5|Body6*	*RUN 5|Body7*	*RUN 5|Body8*	*RUN 5|Body9*	*RUN 5|Body10*	*RUN 5|Body11*
State	Meshed										
Graphics Properties											
Visible	Yes										
Transparency	1										
Definition											

Suppressed	No										
Stiffness Behavior	Flexible										
Coordinate System	Default Coordinate System										
Reference Temperature	By Environment										
Treatment	None										
Material											
Assignment	ECOFLEX-0045										Plastic, ABS (high-impact)
Nonlinear Effects	Yes										
Thermal Strain Effects	Yes										
Bounding Box											
Length X	30. mm	7. mm									4. mm
Length Y	30. mm	7. mm									4. mm
Length Z	1.75 mm	3.5 mm									38. mm
Properties											
Volume	1424.4 mm³	74.166 mm³									608. mm³
Mass	1.5099e-003 kg	7.8616e-005 kg									6.2624e-004 kg
Centroid X	5.8383e-016 mm	-9. mm	1.5275e-010 mm	9. mm	-9. mm	-1.4017e-010 mm	1.3757e-009 mm	9. mm		-9. mm	0. mm
Centroid Y	3.3172e-016 mm	1.0953e-009 mm	1.8953e-010 mm	-1.6294e-010 mm	9. mm	-9. mm	9. mm	-9. mm	9. mm	-9. mm	1.5582e-017 mm
Centroid Z	-0.875 mm	1.4068 mm									22.55 mm
Moment of Inertia Ip1	0.11686 kg·mm²	2.8226e-004 kg·mm²									7.6193e-002 kg·mm²
Moment of Inertia Ip2	0.11686 kg·mm²	2.8226e-004 kg·mm²									7.6193e-002 kg·mm²
Moment of Inertia Ip3	0.23295 kg·mm²	4.4399e-004 kg·mm²									1.67e-003 kg·mm²
Statistics											
Nodes	1763	521	516	517	501	511	503	511	529	496	501
Elements	206	263	256	258	252	254	250	254	268	249	64
Mesh Metric	None										

TABLE 6
Model (B4) > Materials

Object Name	*Materials*
State	Fully Defined
Statistics	
Materials	3
Material Assignments	0

Coordinate Systems

TABLE 7
Model (B4) > Coordinate Systems > Coordinate System

Object Name	*Global Coordinate System*
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. mm
Origin Y	0. mm
Origin Z	0. mm
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]
Transfer Properties	
Source	
Read Only	No

Connections

TABLE 8
Model (B4) > Connections

Object Name	*Connections*
State	Fully Defined
Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	
Enabled	Yes
Statistics	
Contacts	11
Active Contacts	11
Joints	0
Active Joints	0
Beams	0
Active Beams	0
Bearings	0
Active Bearings	0
Springs	0
Active Springs	0
Body Interactions	0
Active Body Interactions	0

TABLE 9
Model (B4) > Connections > Contacts

Object Name	*Contacts*	*Contacts 2*
State	Fully Defined	
Definition		
Connection Type	Contact	
Scope		
Scoping Method	Geometry Selection	
Geometry	All Bodies	
Auto Detection		
Tolerance Type	Slider	
Tolerance Slider	0.	
Tolerance Value	0.15155 mm	
Use Range	No	
Face/Face	Yes	
Face-Face Angle Tolerance	75. °	
Face Overlap Tolerance	Off	
Cylindrical Faces	Include	
Face/Edge	No	
Edge/Edge	No	
Priority	Include All	
Group By	Bodies	
Search Across	Bodies	
Statistics		
Connections	10	1
Active Connections	10	1

TABLE 10
Model (B4) > Connections > Contacts > Contact Regions

Object Name	*Contact Region*	*Contact Region 2*	*Contact Region 3*	*Contact Region 4*	*Contact Region 5*	*Contact Region 6*	*Contact Region 7*	*Contact Region 8*	*Contact Region 9*	*Contact Region 10*
State	Fully Defined									
Scope										
Scoping Method	Geometry Selection									
Contact	1 Face									
Target	1 Face									
Contact Bodies	RUN 5|Body1									RUN 5|Body3
Target Bodies	RUN 5|Body2	RUN 5|Body3	RUN 5|Body4	RUN 5|Body5	RUN 5|Body6	RUN 5|Body7	RUN 5|Body8	RUN 5|Body9	RUN 5|Body10	RUN 5|Body11
Protected	No									
Definition										
Type	Bonded									
Scope Mode	Automatic									
Behavior	Program Controlled									
Trim Contact	Program Controlled									
Trim Tolerance	0.15155 mm									
Contact APDL Name										

Target APDL Name	
Suppressed	No
Display	
Element Normals	No
Advanced	
Formulation	Program Controlled
Small Sliding	Program Controlled
Detection Method	Program Controlled
Penetration Tolerance	Program Controlled
Elastic Slip Tolerance	Program Controlled
Normal Stiffness	Program Controlled
Update Stiffness	Program Controlled
Pinball Region	Program Controlled
Geometric Modification	
Contact Geometry Correction	None
Target Geometry Correction	None

TABLE 11
Model (B4) > Connections > Contacts 2 > Contact Regions

Object Name	*Frictionless - RUN 5|Body3 To RUN 5|Body11*
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Contact	1 Face
Target	1 Face
Contact Bodies	RUN 5|Body3
Target Bodies	RUN 5|Body11
Protected	No
Definition	
Type	Frictionless
Scope Mode	Manual
Behavior	Asymmetric
Trim Contact	Program Controlled
Contact APDL Name	
Target APDL Name	
Suppressed	No
Display	
Element Normals	No
Advanced	
Formulation	Augmented Lagrange
Small Sliding	Program Controlled
Detection Method	Program Controlled
Penetration Tolerance	Program Controlled
Normal Stiffness	Program Controlled
Update Stiffness	Program Controlled
Stabilization Damping Factor	0.
Pinball Region	Program Controlled
Time Step Controls	None
Geometric Modification	
Interface Treatment	Add Offset, No Ramping
Offset	0. mm
Contact Geometry Correction	None
Target Geometry Correction	None

Mesh

TABLE 12
Model (B4) > Mesh

Object Name	*Mesh*
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled

Element Size	Default
Sizing	
Use Adaptive Sizing	Yes
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	60.621 mm
Average Surface Area	67.026 mm²
Minimum Edge Length	0.45 mm
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5.e-002)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Auto-Map Fillets	No
Automatic Methods	
Sheet Body Method	Prime Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	6869
Elements	2574
Show Detailed Statistics	No

Static Structural (B5)

TABLE 13
Model (B4) > Analysis

Object Name	*Static Structural (B5)*
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 14
Model (B4) > Static Structural (B5) > Analysis Settings

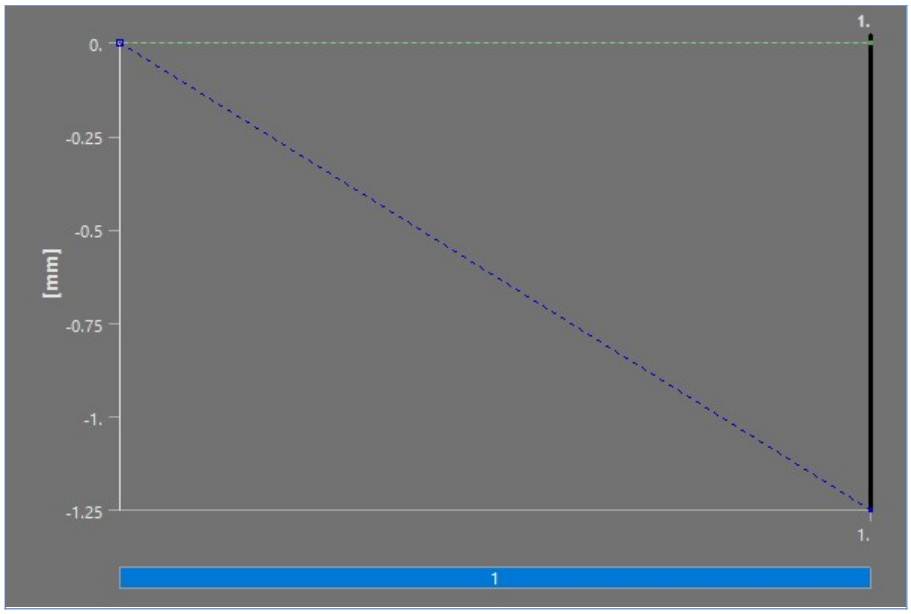
Object Name	*Analysis Settings*
State	Fully Defined
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	On
Define By	Substeps
Initial Substeps	20.
Minimum Substeps	5.

Maximum Substeps	500.
Solver Controls	
Solver Type	Program Controlled
Weak Springs	On
Spring Stiffness	Program Controlled
Solver Pivot Checking	Program Controlled
Large Deflection	On
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\DLSU\Downloads\ANSYS FOR HOLLOW RODS\PROJECT 2_files\dp0\SYS-1\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	Yes
Solver Units	Active System
Solver Unit System	nmm

TABLE 15
Model (B4) > Static Structural (B5) > Loads

Object Name	*Fixed Support*	*Displacement*
State	Fully Defined	
Scope		
Scoping Method	Geometry Selection	
Geometry	1 Face	
Definition		
Type	Fixed Support	Displacement
Suppressed	No	
Define By		Components
Coordinate System		Global Coordinate System
X Component		0. mm (ramped)
Y Component		0. mm (ramped)
Z Component		-1.25 mm (ramped)

FIGURE 1
Model (B4) > Static Structural (B5) > Displacement



Solution (B6)

TABLE 16
Model (B4) > Static Structural (B5) > Solution

Object Name	*Solution (B6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	26. s
MAPDL Memory Used	241. MB
MAPDL Result File Size	12.562 MB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 17
Model (B4) > Static Structural (B5) > Solution (B6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 18
Model (B4) > Static Structural (B5) > Solution (B6) > Results

Object Name	*Total Deformation*	*Directional Deformation*	*Directional Deformation 2*	*Directional Deformation 3*	*Equivalent Elastic Strain*	*Maximum Principal Elastic Strain*	*Maximum Shear Elastic Strain*	*Elastic Strain Intensity*	*Equivalent Stress*	*Maximum Principal Stress*	*Maximum Shear Stress*
State	Solved										
Scope											
Scoping	Geometry Selection										

Method										
Geometry	2 Bodies									
Definition										
Type	Total Deformation	Directional Deformation			Equivalent Elastic Strain	Maximum Principal Elastic Strain	Maximum Shear Elastic Strain	Elastic Strain Intensity	Equivalent (von-Mises) Stress	Maximum Principal Stress	Maximum Shear Stress
By	Time									
Display Time	Last									
Separate Data by Entity	No									
Calculate Time History	Yes									
Identifier										
Suppressed	No									
Orientation		X Axis	Y Axis	Z Axis						
Coordinate System		Global Coordinate System								
Results										
Minimum	0. mm	-0.31704 mm	-0.35807 mm	-1.2772 mm	1.3956e-005 mm/mm	9.7158e-006 mm/mm	2.3236e-005 mm/mm	1.9803e-006 MPa	-2.9198e-002 MPa	1.099e-006 MPa
Maximum	1.2784 mm	0.34379 mm	0.34683 mm	5.7538e-002 mm	0.39425 mm/mm	0.22387 mm/mm	0.5966 mm/mm	4.7683e-002 MPa	2.6837e-002 MPa	2.5752e-002 MPa
Average	9.738e-002 mm	-9.2917e-004 mm	-5.0712e-004 mm	-6.6751e-002 mm	5.1909e-002 mm/mm	2.9012e-002 mm/mm	7.4378e-002 mm/mm	6.1699e-003 MPa	3.603e-004 MPa	3.3382e-003 MPa
Minimum Occurs On	RUN 5|Body1	RUN 5|Body3			RUN 5|Body1				RUN 5|Body3	RUN 5|Body1
Maximum Occurs On	RUN 5|Body3			RUN 5|Body1	RUN 5|Body3					
Minimum Value Over Time										
Minimum	0. mm	-0.31704 mm	-0.35807 mm	-1.2772 mm	6.169e-007 mm/mm	4.291e-007 mm/mm	1.0269e-006 mm/mm	8.7538e-008 MPa	-3.0771e-002 MPa	4.857e-008 MPa
Maximum	0. mm	-1.3886e-002 mm	-1.5501e-002 mm	-6.7546e-002 mm	1.3956e-005 mm/mm	9.7158e-006 mm/mm	2.3236e-005 mm/mm	1.9803e-006 MPa	-2.5231e-003 MPa	1.099e-006 MPa
Maximum Value Over Time										
Minimum	6.76e-002 mm	1.3364e-002 mm	1.4178e-002 mm	2.3949e-003 mm	1.9438e-002 mm/mm	1.3041e-002 mm/mm	3.1041e-002 mm/mm	2.6194e-003 MPa	9.1802e-004 MPa	1.4611e-003 MPa
Maximum	1.2784 mm	0.34379 mm	0.34683 mm	5.7538e-002 mm	0.39425 mm/mm	0.22387 mm/mm	0.5966 mm/mm	4.7683e-002 MPa	2.6837e-002 MPa	2.5752e-002 MPa
Information										
Time	1. s									
Load Step	1									
Substep	8									
Iteration Number	16									
Integration Point Results										
Display Option					Averaged					
Average Across Bodies					No					

FIGURE 2
Model (B4) > Static Structural (B5) > Solution (B6) > Total Deformation

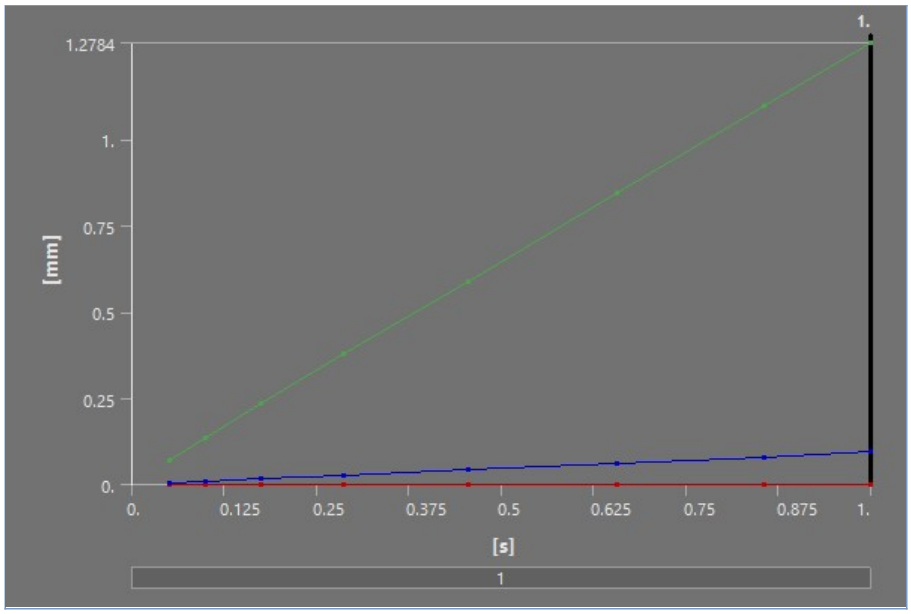


TABLE 19
Model (B4) > Static Structural (B5) > Solution (B6) > Total Deformation

Time [s]	Minimum [mm]	Maximum [mm]	Average [mm]
5.e-002	0.	6.76e-002	4.6442e-003
0.1		0.13445	9.2921e-003
0.175		0.23322	1.6267e-002
0.2875		0.37774	2.6724e-002
0.45625		0.58706	4.2309e-002
0.65625		0.84218	6.047e-002
0.85625		1.0971	7.9953e-002
1.		1.2784	9.738e-002

FIGURE 3
Model (B4) > Static Structural (B5) > Solution (B6) > Total Deformation > Figure

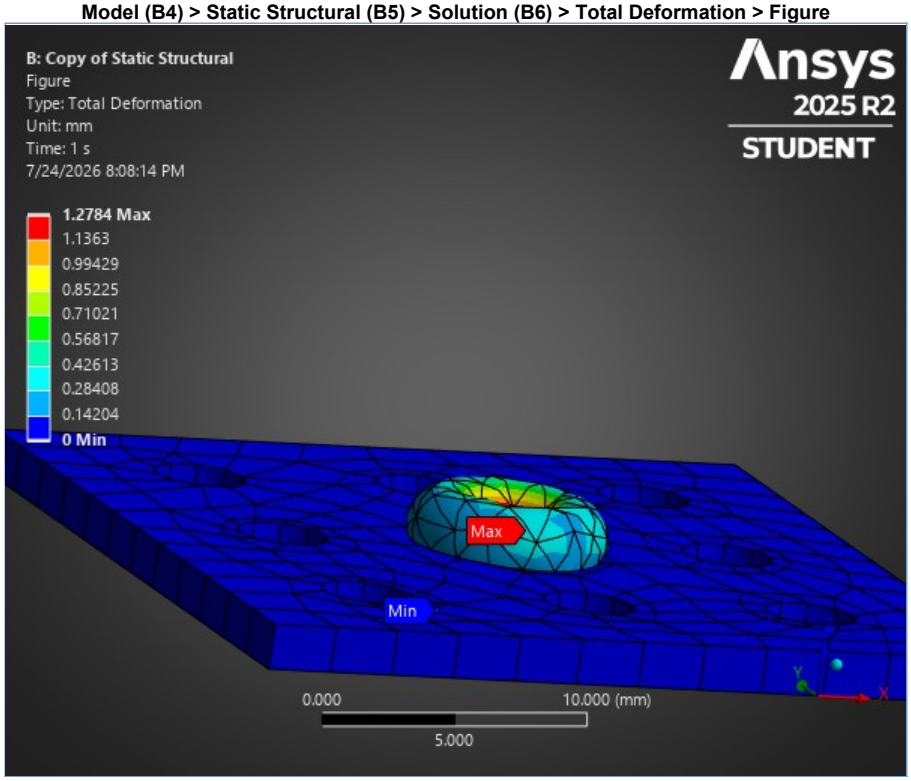


FIGURE 4
Model (B4) > Static Structural (B5) > Solution (B6) > Directional Deformation

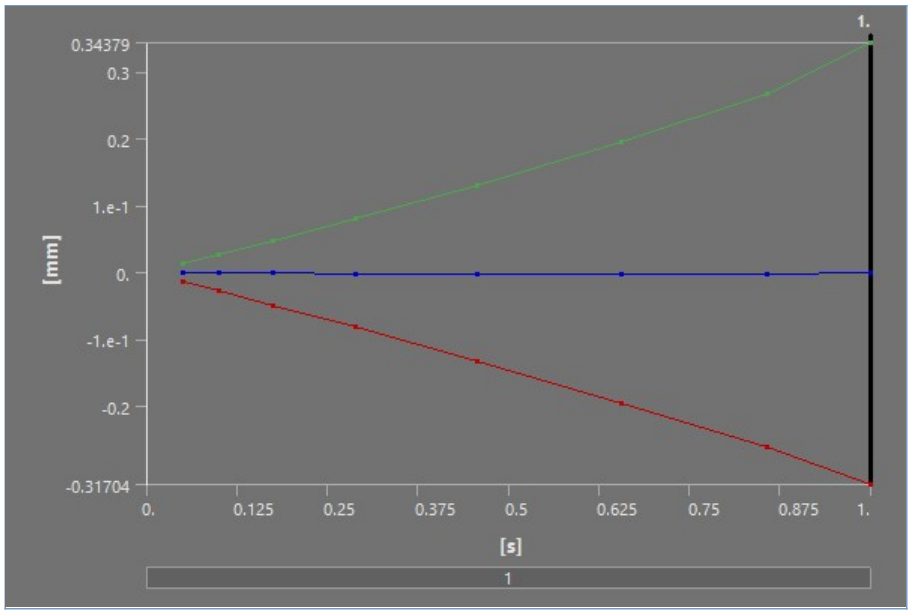


TABLE 20
Model (B4) > Static Structural (B5) > Solution (B6) > Directional Deformation

Time [s]	Minimum [mm]	Maximum [mm]	Average [mm]
5.e-002	-1.3886e-002	1.3364e-002	-2.7665e-004
0.1	-2.7951e-002	2.6974e-002	-5.5033e-004
0.175	-4.9368e-002	4.7852e-002	-9.5412e-004
0.2875	-8.2123e-002	8.0189e-002	-1.5395e-003
0.45625	-0.13257	0.13122	-2.3614e-003
0.65625	-0.19572	0.19507	-3.1798e-003
0.85625	-0.26096	0.26838	-2.6015e-003
1.	-0.31704	0.34379	-9.2917e-004

FIGURE 5
Model (B4) > Static Structural (B5) > Solution (B6) > Directional Deformation > Figure

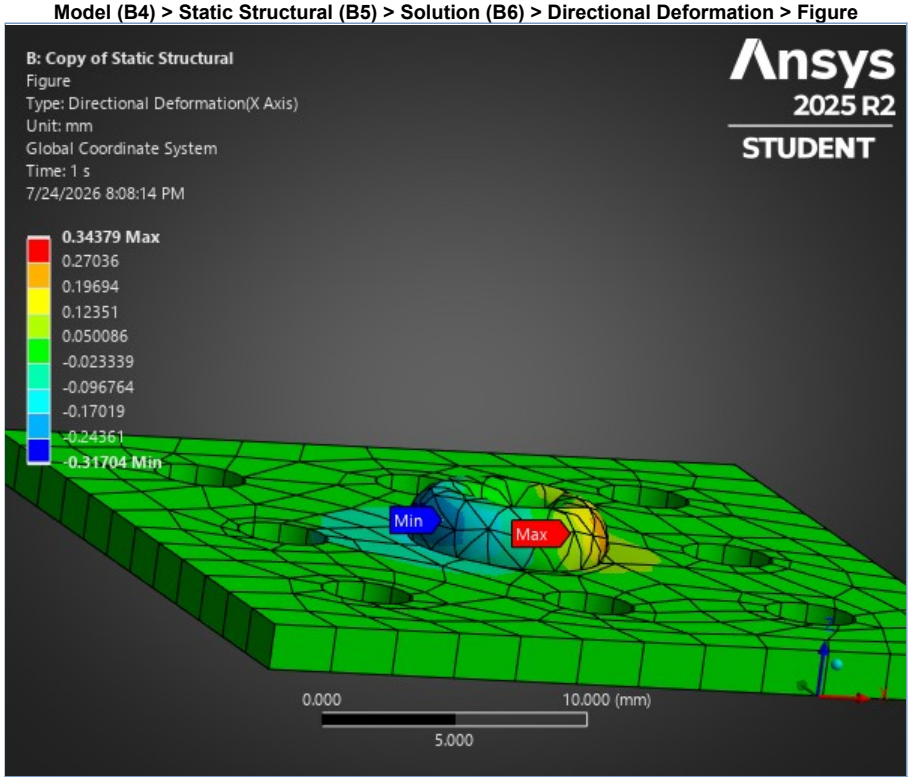


FIGURE 6
Model (B4) > Static Structural (B5) > Solution (B6) > Directional Deformation 2

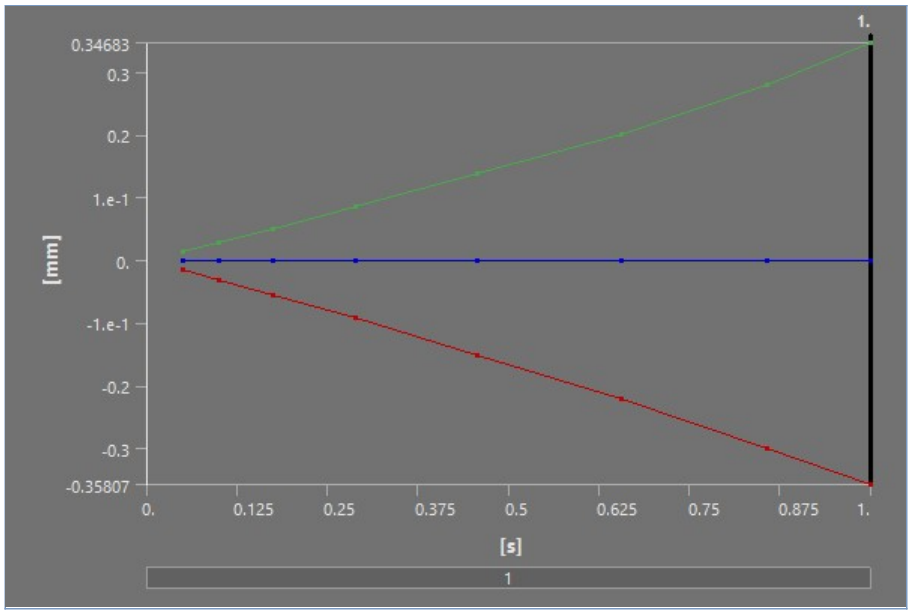


TABLE 21
Model (B4) > Static Structural (B5) > Solution (B6) > Directional Deformation 2

Time [s]	Minimum [mm]	Maximum [mm]	Average [mm]
5.e-002	-1.5501e-002	1.4178e-002	-5.7781e-005
0.1	-3.1278e-002	2.8588e-002	-1.1624e-004
0.175	-5.5446e-002	5.0623e-002	-2.0443e-004
0.2875	-9.2771e-002	8.4549e-002	-3.3596e-004
0.45625	-0.15085	0.13729	-5.0684e-004
0.65625	-0.22202	0.2017	-6.4375e-004
0.85625	-0.2999	0.27943	-9.6001e-004
1.	-0.35807	0.34683	-5.0712e-004

FIGURE 7
Model (B4) > Static Structural (B5) > Solution (B6) > Directional Deformation 2 > Figure

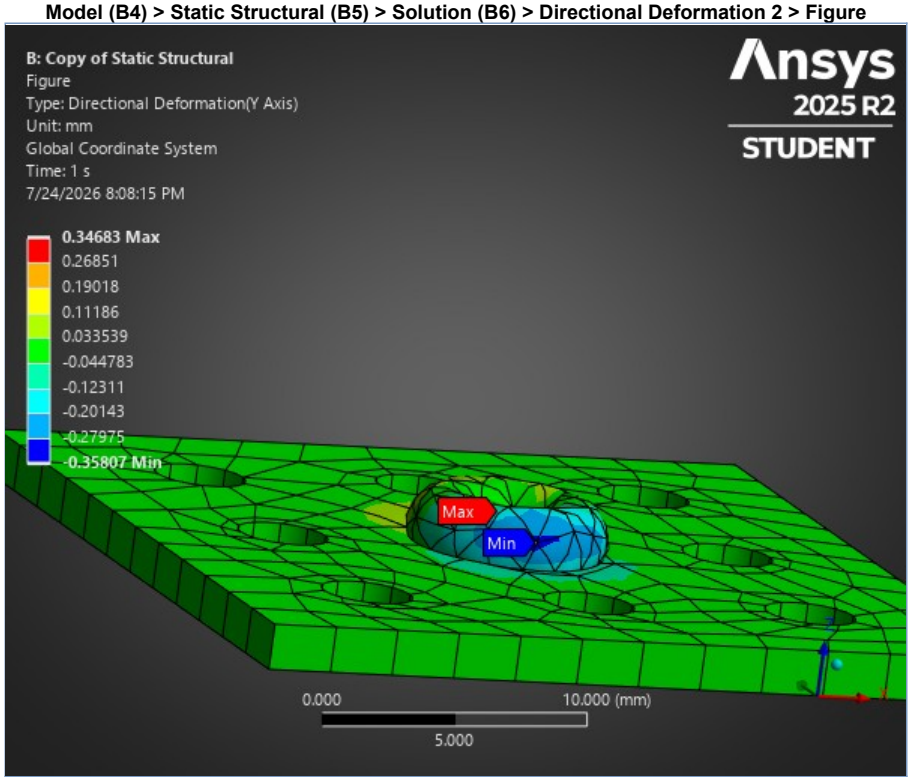


FIGURE 8
Model (B4) > Static Structural (B5) > Solution (B6) > Directional Deformation 3

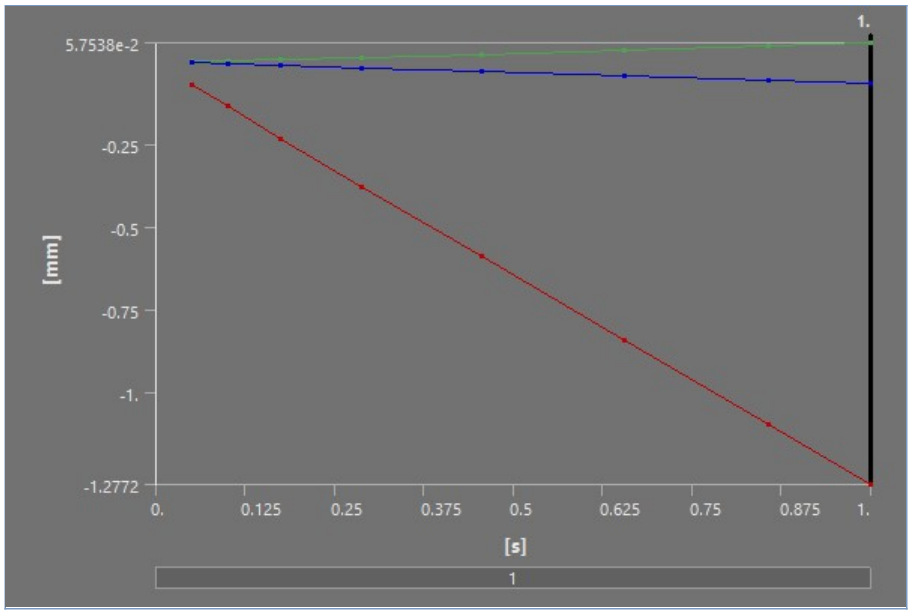


TABLE 22
Model (B4) > Static Structural (B5) > Solution (B6) > Directional Deformation 3

Time [s]	Minimum [mm]	Maximum [mm]	Average [mm]
5.e-002	-6.7546e-002	2.3949e-003	-3.4991e-003
0.1	-0.13434	4.8318e-003	-6.9695e-003
0.175	-0.23303	8.5655e-003	-1.2116e-002
0.2875	-0.37744	1.4342e-002	-1.9685e-002
0.45625	-0.58669	2.3353e-002	-3.0589e-002
0.65625	-0.84155	3.4495e-002	-4.2655e-002
0.85625	-1.0959	4.6474e-002	-5.5255e-002
1.	-1.2772	5.7538e-002	-6.6751e-002

FIGURE 9
Model (B4) > Static Structural (B5) > Solution (B6) > Directional Deformation 3 > Figure

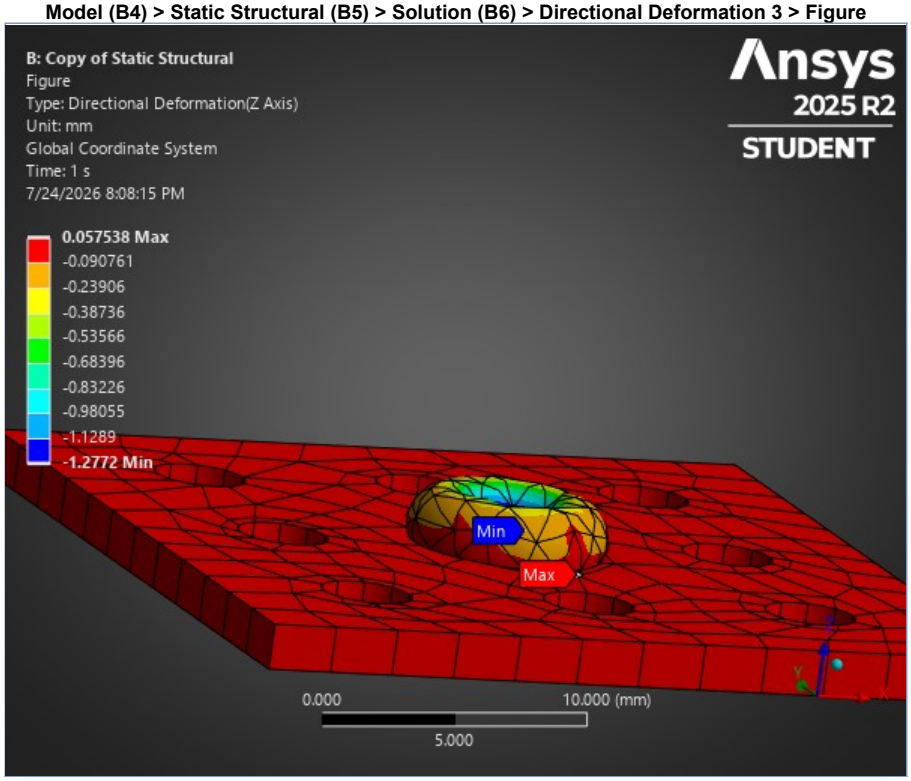


FIGURE 10
Model (B4) > Static Structural (B5) > Solution (B6) > Equivalent Elastic Strain

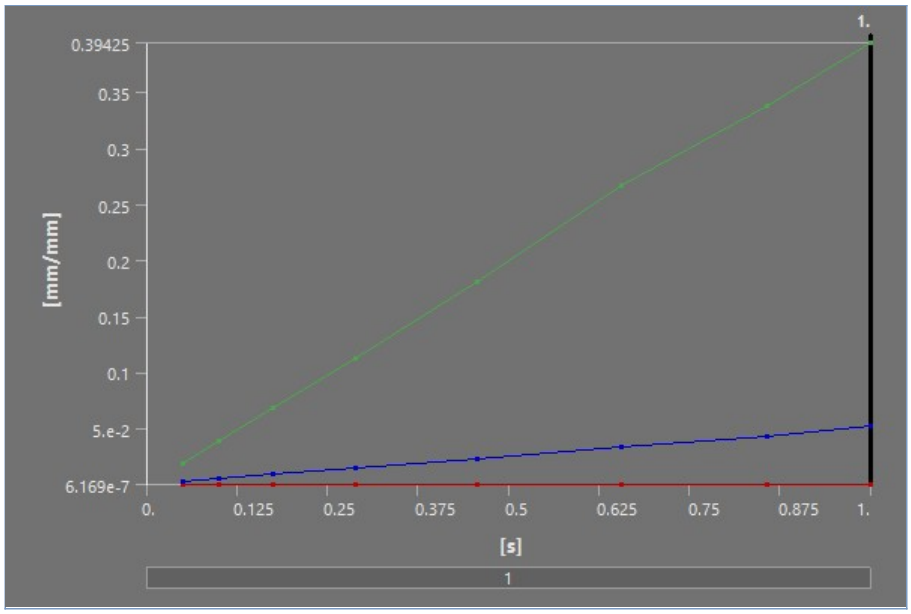


TABLE 23
Model (B4) > Static Structural (B5) > Solution (B6) > Equivalent Elastic Strain

Time [s]	Minimum [mm/mm]	Maximum [mm/mm]	Average [mm/mm]
5.e-002	6.169e-007	1.9438e-002	2.4821e-003
0.1	1.2386e-006	3.8947e-002	4.9729e-003
0.175	2.1793e-006	6.8381e-002	8.7233e-003
0.2875	3.6051e-006	0.11304	1.4373e-002
0.45625	5.7523e-006	0.1816	2.2865e-002
0.65625	8.2627e-006	0.26628	3.2879e-002
0.85625	1.1087e-005	0.33759	4.3299e-002
1.	1.3956e-005	0.39425	5.1909e-002

FIGURE 11
Model (B4) > Static Structural (B5) > Solution (B6) > Equivalent Elastic Strain > Figure

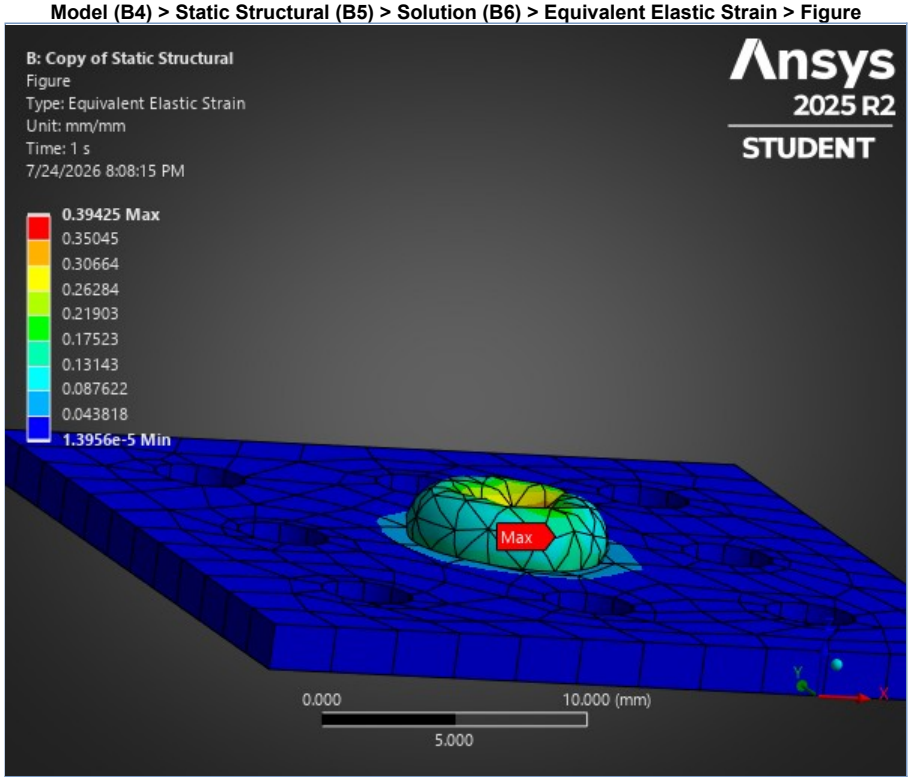


FIGURE 12
Model (B4) > Static Structural (B5) > Solution (B6) > Maximum Principal Elastic Strain

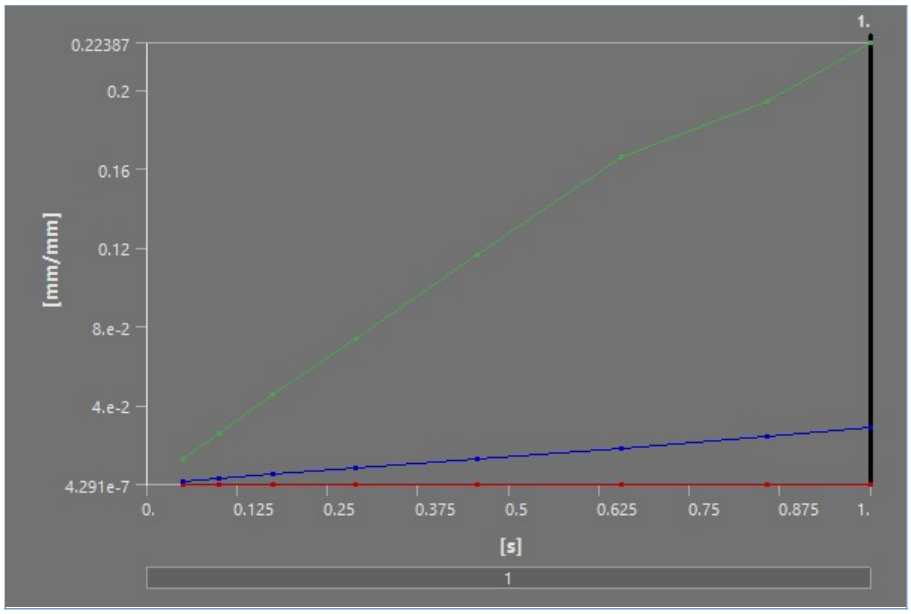


TABLE 24
Model (B4) > Static Structural (B5) > Solution (B6) > Maximum Principal Elastic Strain

Time [s]	Minimum [mm/mm]	Maximum [mm/mm]	Average [mm/mm]
5.e-002	4.291e-007	1.3041e-002	1.4187e-003
0.1	8.6146e-007	2.6021e-002	2.8407e-003
0.175	1.5154e-006	4.5378e-002	4.9775e-003
0.2875	2.506e-006	7.4156e-002	8.1839e-003
0.45625	3.9962e-006	0.1167	1.2959e-002
0.65625	5.7349e-006	0.16607	1.8487e-002
0.85625	7.6973e-006	0.19449	2.4188e-002
1.	9.7158e-006	0.22387	2.9012e-002

FIGURE 13
Model (B4) > Static Structural (B5) > Solution (B6) > Maximum Principal Elastic Strain > Figure

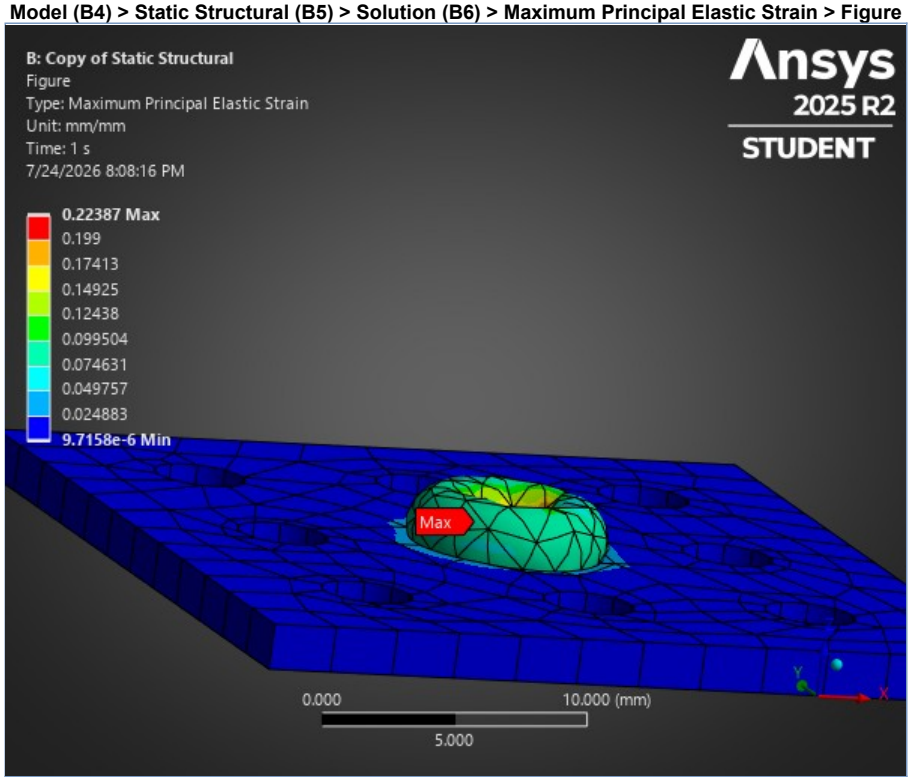


FIGURE 14
Model (B4) > Static Structural (B5) > Solution (B6) > Maximum Shear Elastic Strain

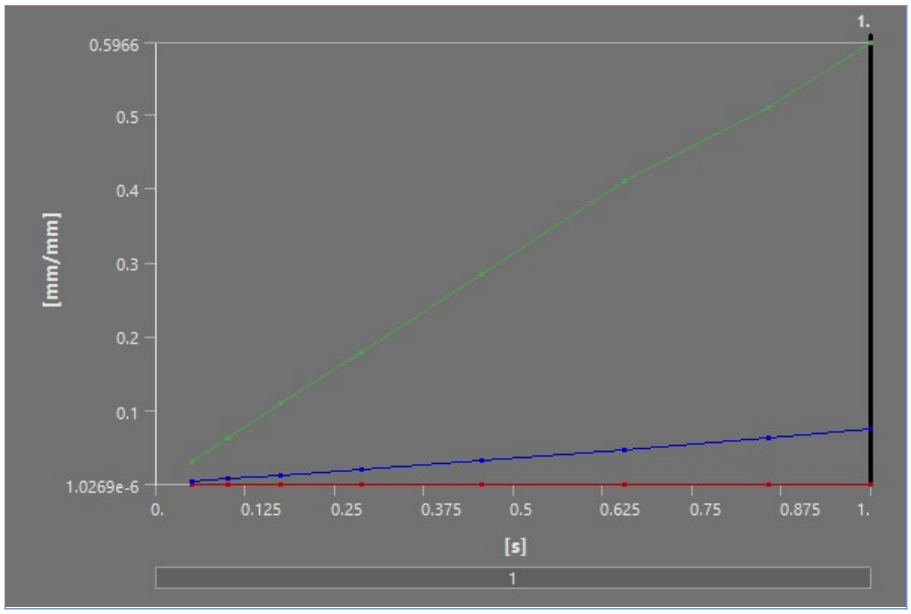


TABLE 25
Model (B4) > Static Structural (B5) > Solution (B6) > Maximum Shear Elastic Strain

Time [s]	Minimum [mm/mm]	Maximum [mm/mm]	Average [mm/mm]
5.e-002	1.0269e-006	3.1041e-002	3.6848e-003
0.1	2.0617e-006	6.2059e-002	7.3735e-003
0.175	3.6271e-006	0.10857	1.2907e-002
0.2875	5.9998e-006	0.1784	2.119e-002
0.45625	9.5717e-006	0.28348	3.3466e-002
0.65625	1.3745e-005	0.40905	4.7566e-002
0.85625	1.8445e-005	0.5089	6.2046e-002
1.	2.3236e-005	0.5966	7.4378e-002

FIGURE 15
Model (B4) > Static Structural (B5) > Solution (B6) > Maximum Shear Elastic Strain > Figure

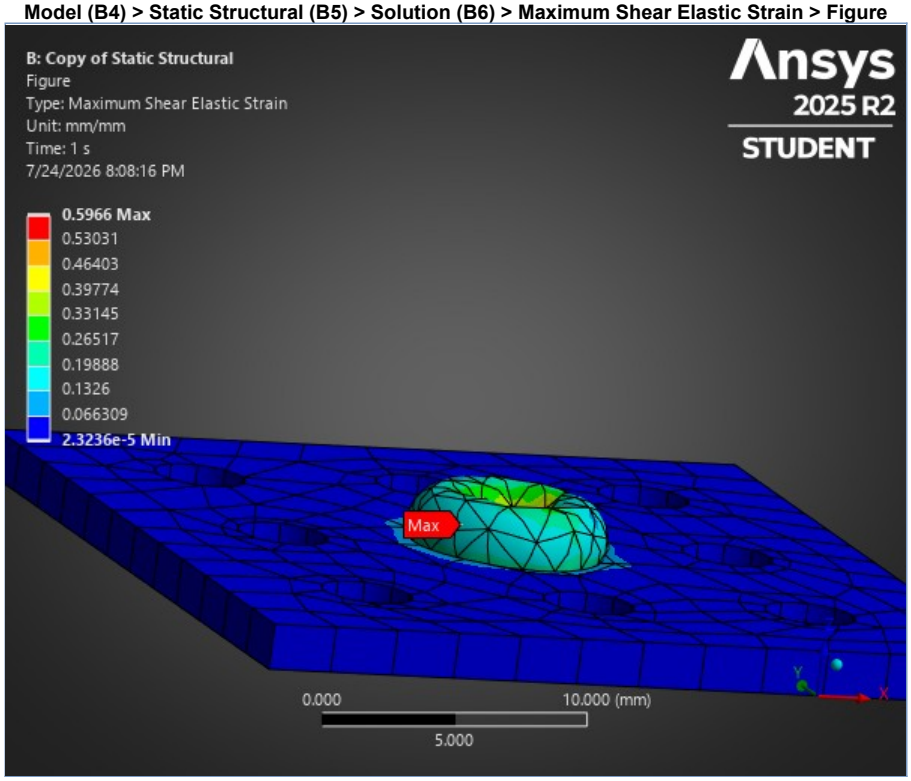


FIGURE 16
Model (B4) > Static Structural (B5) > Solution (B6) > Elastic Strain Intensity

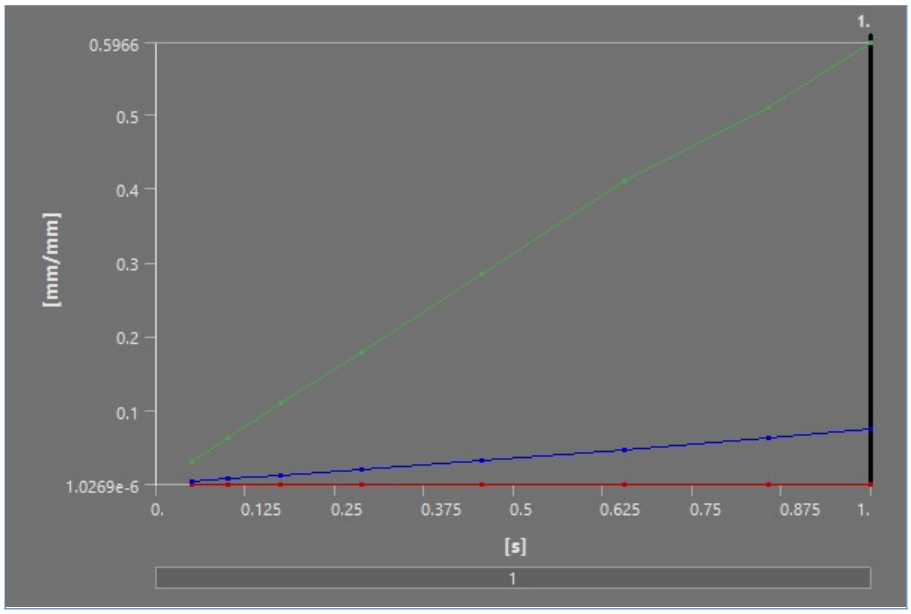


TABLE 26
Model (B4) > Static Structural (B5) > Solution (B6) > Elastic Strain Intensity

Time [s]	Minimum [mm/mm]	Maximum [mm/mm]	Average [mm/mm]
5.e-002	1.0269e-006	3.1041e-002	3.6848e-003
0.1	2.0617e-006	6.2059e-002	7.3735e-003
0.175	3.6271e-006	0.10857	1.2907e-002
0.2875	5.9998e-006	0.1784	2.119e-002
0.45625	9.5717e-006	0.28348	3.3466e-002
0.65625	1.3745e-005	0.40905	4.7566e-002
0.85625	1.8445e-005	0.5089	6.2046e-002
1.	2.3236e-005	0.5966	7.4378e-002

FIGURE 17
Model (B4) > Static Structural (B5) > Solution (B6) > Elastic Strain Intensity > Figure

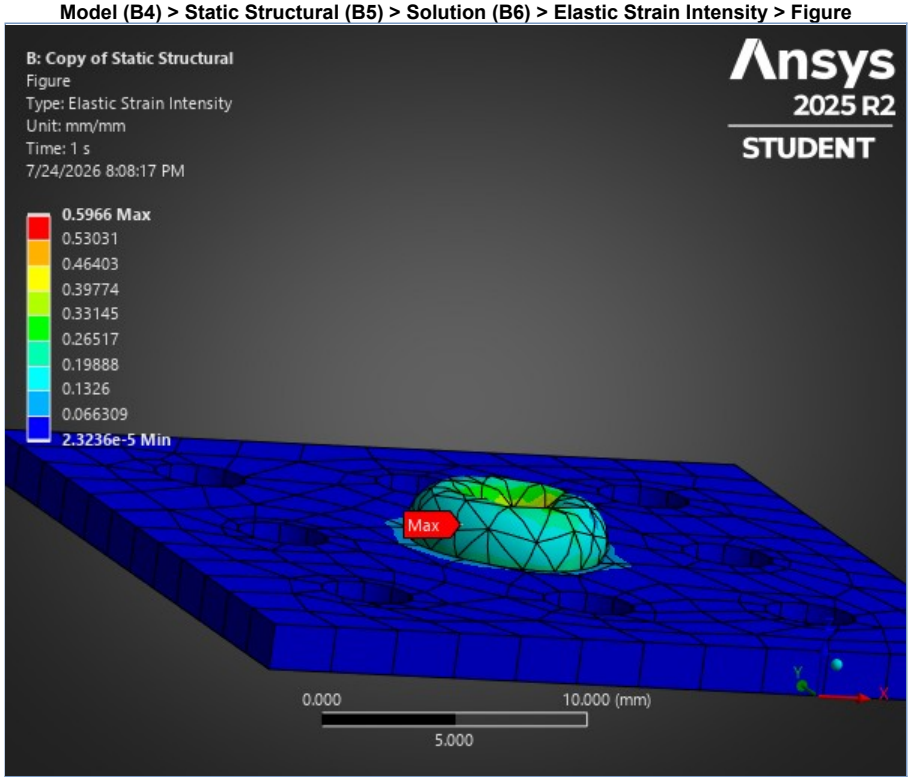


FIGURE 18
Model (B4) > Static Structural (B5) > Solution (B6) > Equivalent Stress

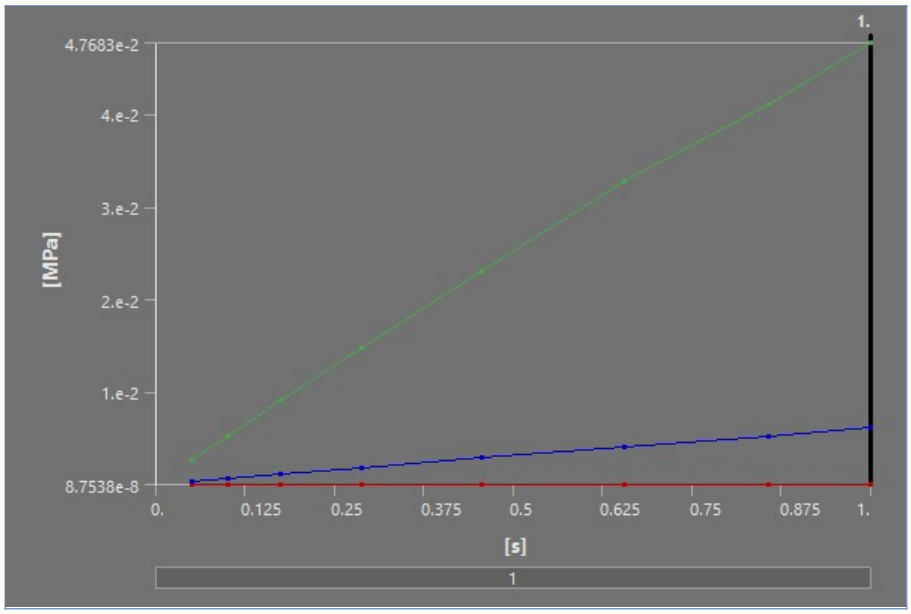


TABLE 27
Model (B4) > Static Structural (B5) > Solution (B6) > Equivalent Stress

Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
5.e-002	8.7538e-008	2.6194e-003	3.2452e-004
0.1	1.7576e-007	5.2074e-003	6.4653e-004
0.175	3.0924e-007	9.0384e-003	1.1246e-003
0.2875	5.1156e-007	1.4693e-002	1.83e-003
0.45625	8.1625e-007	2.3032e-002	2.8564e-003
0.65625	1.1725e-006	3.2815e-002	4.0135e-003
0.85625	1.5732e-006	4.1026e-002	5.1828e-003
1.	1.9803e-006	4.7683e-002	6.1699e-003

FIGURE 19
Model (B4) > Static Structural (B5) > Solution (B6) > Equivalent Stress > Figure

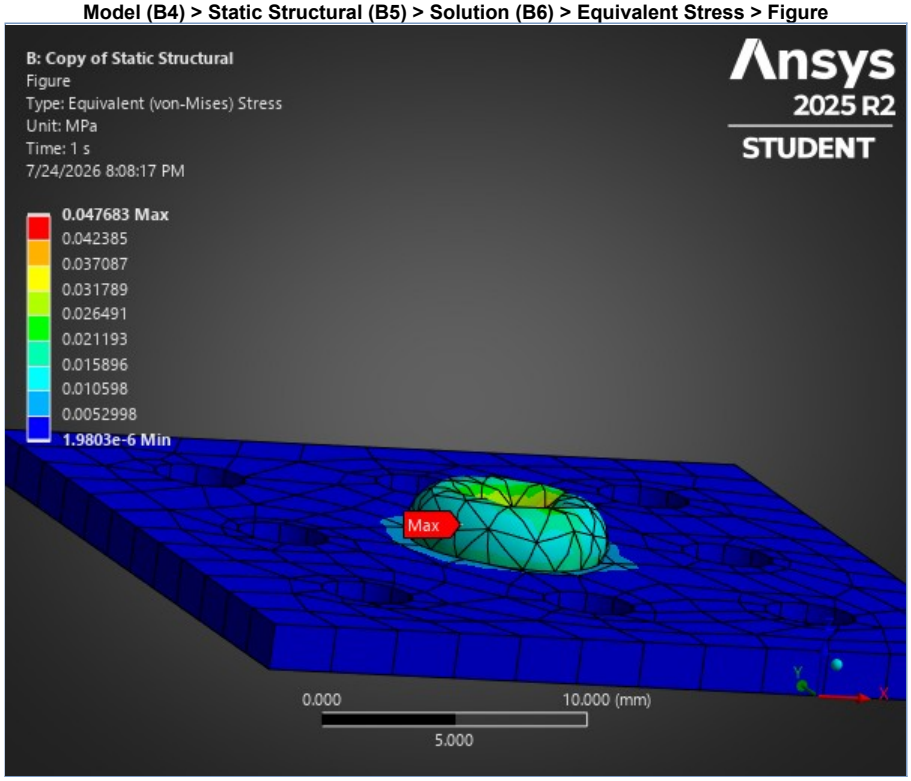


FIGURE 20
Model (B4) > Static Structural (B5) > Solution (B6) > Maximum Principal Stress

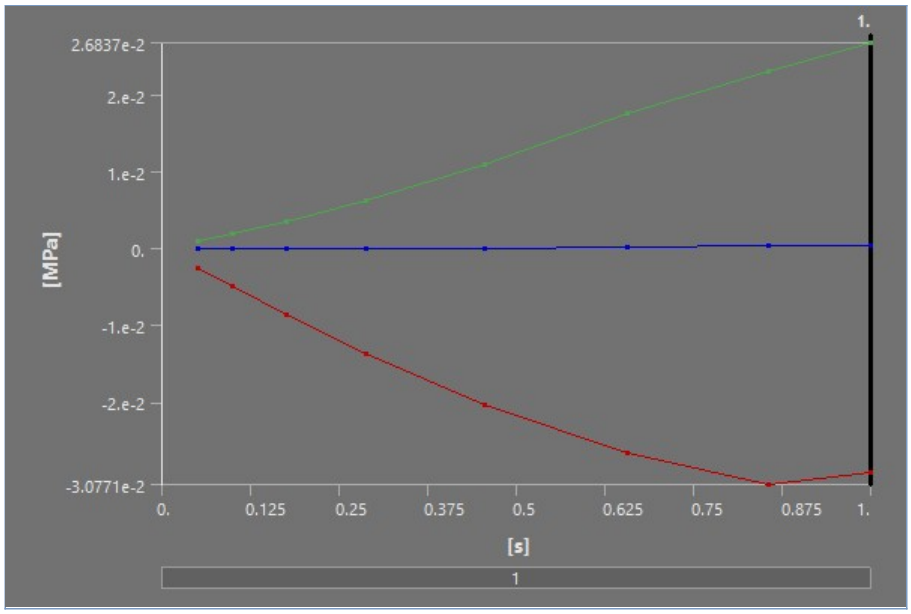


TABLE 28
Model (B4) > Static Structural (B5) > Solution (B6) > Maximum Principal Stress

Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
5.e-002	-2.5231e-003	9.1802e-004	-1.0013e-005
0.1	-4.9921e-003	1.9007e-003	-1.6872e-005
0.175	-8.5811e-003	3.4998e-003	-2.0898e-005
0.2875	-1.3661e-002	6.1885e-003	-1.1828e-005
0.45625	-2.043e-002	1.0919e-002	4.0148e-005
0.65625	-2.6721e-002	1.7621e-002	1.6804e-004
0.85625	-3.0771e-002	2.308e-002	3.251e-004
1.	-2.9198e-002	2.6837e-002	3.603e-004

FIGURE 21
Model (B4) > Static Structural (B5) > Solution (B6) > Maximum Principal Stress > Figure

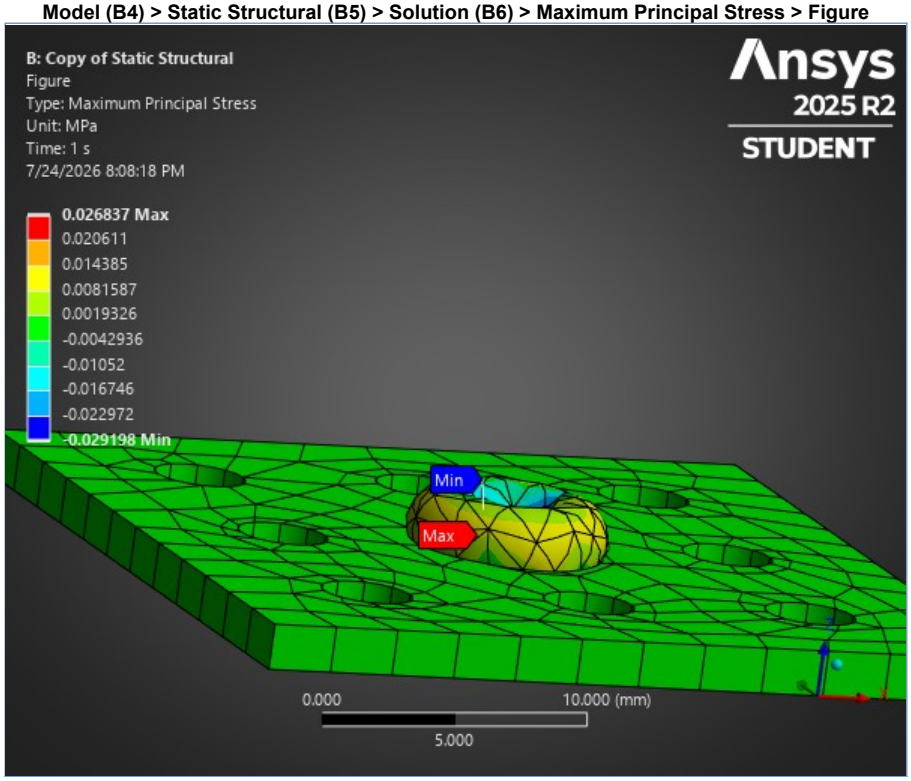


FIGURE 22
Model (B4) > Static Structural (B5) > Solution (B6) > Maximum Shear Stress

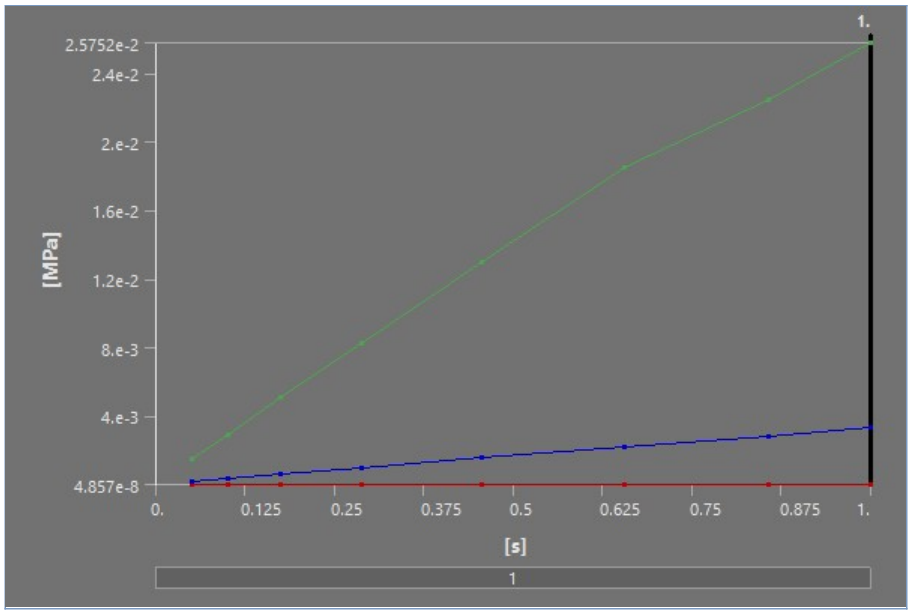


TABLE 29
Model (B4) > Static Structural (B5) > Solution (B6) > Maximum Shear Stress

Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
5.e-002	4.857e-008	1.4611e-003	1.7364e-004
0.1	9.7517e-008	2.9075e-003	3.462e-004
0.175	1.7156e-007	5.0534e-003	6.0289e-004
0.2875	2.8379e-007	8.2299e-003	9.8262e-004
0.45625	4.5274e-007	1.2929e-002	1.5372e-003
0.65625	6.5015e-007	1.8453e-002	2.1646e-003
0.85625	8.7243e-007	2.239e-002	2.8005e-003
1.	1.099e-006	2.5752e-002	3.3382e-003

FIGURE 23
Model (B4) > Static Structural (B5) > Solution (B6) > Maximum Shear Stress > Figure

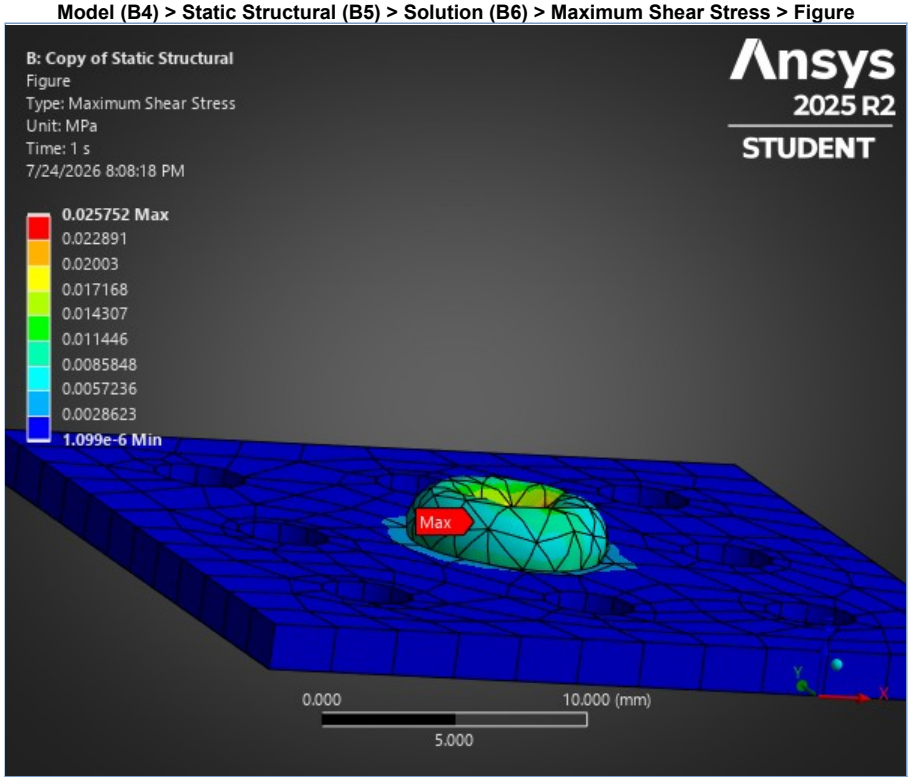


TABLE 30
Model (B4) > Static Structural (B5) > Solution (B6) > Results

Object Name	*Stress Intensity*	*Strain Energy*
State	Solved	
Scope		
Scoping Method	Geometry Selection	
Geometry	2 Bodies	
Definition		
Type	Stress Intensity	Strain Energy
By	Time	
Display Time	Last	
Separate Data by Entity	No	
Calculate Time History	Yes	
Identifier		
Suppressed	No	
Integration Point Results		
Display Option	Averaged	
Average Across Bodies	No	
Results		
Minimum	2.1981e-006 MPa	3.9475e-009 mJ
Maximum	5.1504e-002 MPa	7.1484e-003 mJ
Average	6.6763e-003 MPa	
Minimum Occurs On	RUN 5|Body1	
Maximum Occurs On	RUN 5|Body3	
Total		0.31364 mJ
Minimum Value Over Time		
Minimum	9.714e-008 MPa	7.348e-012 mJ
Maximum	2.1981e-006 MPa	3.9475e-009 mJ
Maximum Value Over Time		
Minimum	2.9221e-003 MPa	1.8866e-005 mJ
Maximum	5.1504e-002 MPa	7.1484e-003 mJ
Information		
Time	1. s	
Load Step	1	
Substep	8	
Iteration Number	16	

FIGURE 24
Model (B4) > Static Structural (B5) > Solution (B6) > Stress Intensity

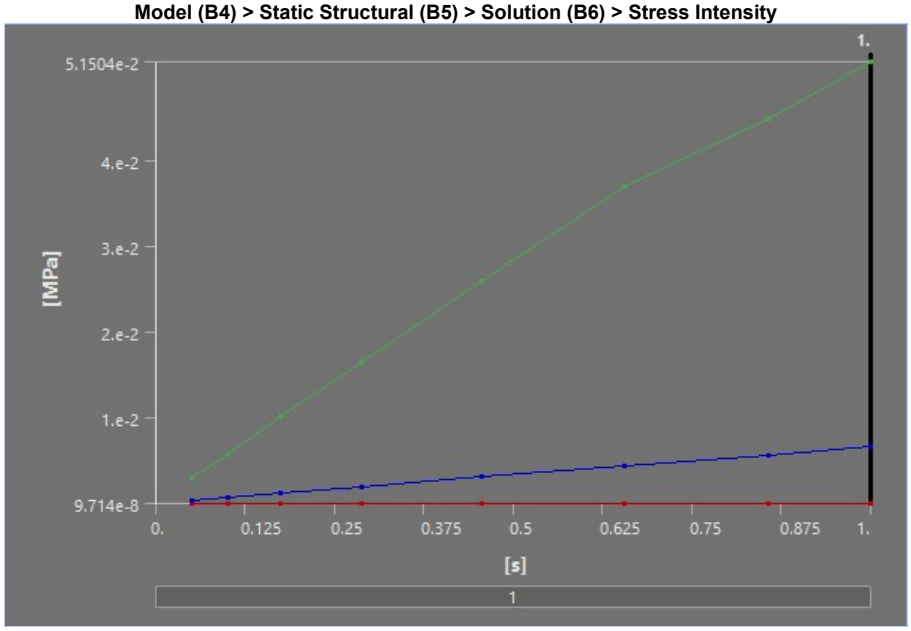


TABLE 31
Model (B4) > Static Structural (B5) > Solution (B6) > Stress Intensity

Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
5.e-002	9.714e-008	2.9221e-003	3.4728e-004
0.1	1.9503e-007	5.815e-003	6.924e-004
0.175	3.4313e-007	1.0107e-002	1.2058e-003
0.2875	5.6758e-007	1.646e-002	1.9652e-003
0.45625	9.0548e-007	2.5857e-002	3.0743e-003
0.65625	1.3003e-006	3.6907e-002	4.3293e-003

0.85625	1.7449e-006	4.478e-002	5.6009e-003
1.	2.1981e-006	5.1504e-002	6.6763e-003

FIGURE 25
Model (B4) > Static Structural (B5) > Solution (B6) > Stress Intensity > Figure

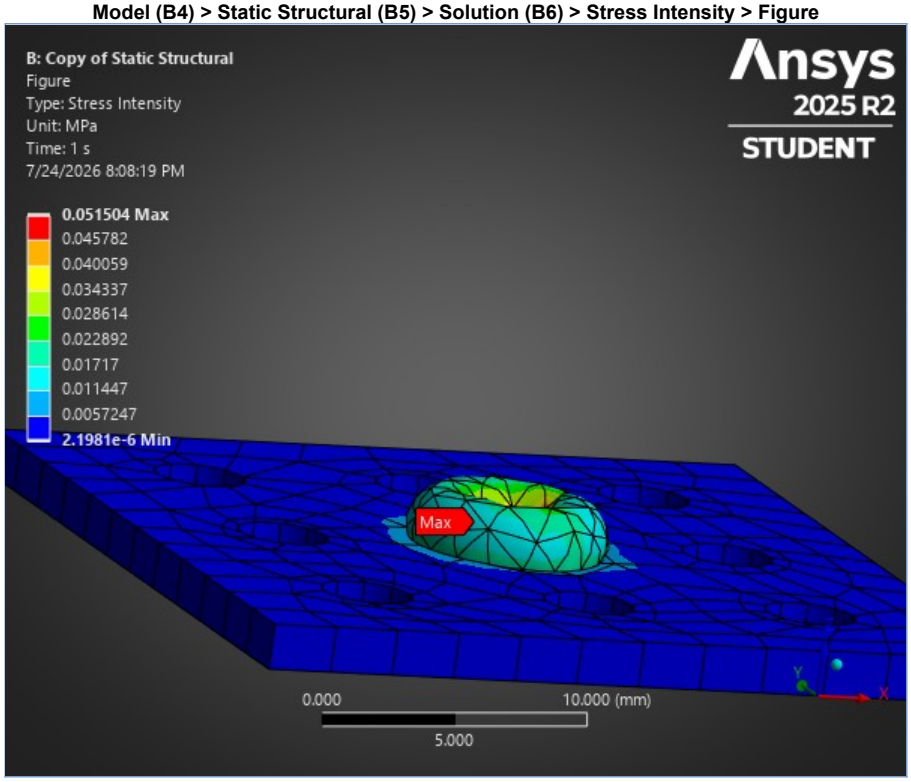


FIGURE 26
Model (B4) > Static Structural (B5) > Solution (B6) > Strain Energy

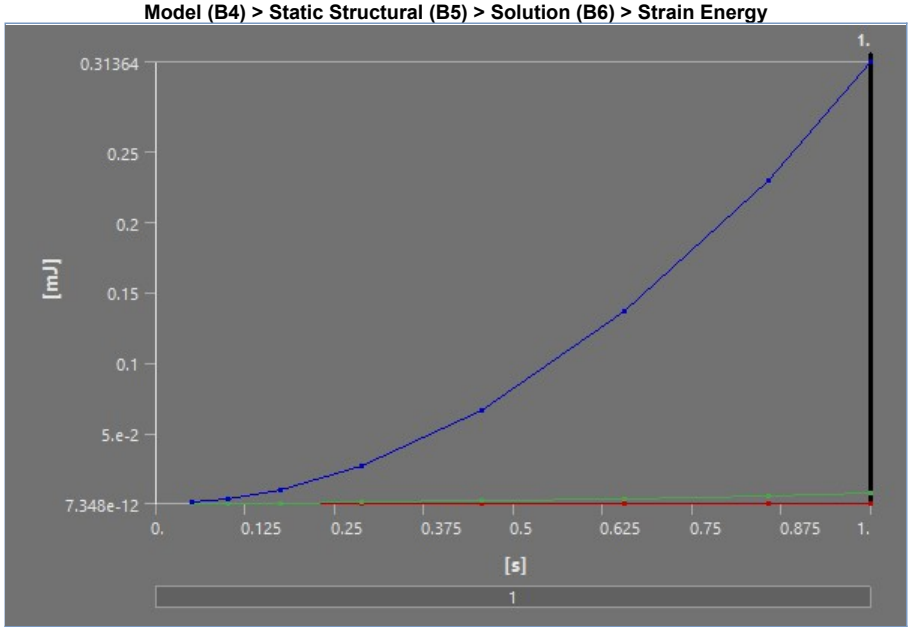


TABLE 32
Model (B4) > Static Structural (B5) > Solution (B6) > Strain Energy

Time [s]	Minimum [mJ]	Maximum [mJ]	Total [mJ]
5.e-002	7.348e-012	1.8866e-005	7.9703e-004
0.1	2.982e-011	7.5747e-005	3.1904e-003
0.175	9.3262e-011	2.3325e-004	9.7772e-003
0.2875	2.5872e-010	6.3444e-004	2.639e-002
0.45625	6.5887e-010	1.6132e-003	6.6297e-002
0.65625	1.3591e-009	3.3516e-003	0.13608

0.85625	2.4622e-009	5.565e-003	0.22922
1.	3.9475e-009	7.1484e-003	0.31364

FIGURE 27
Model (B4) > Static Structural (B5) > Solution (B6) > Strain Energy > Figure

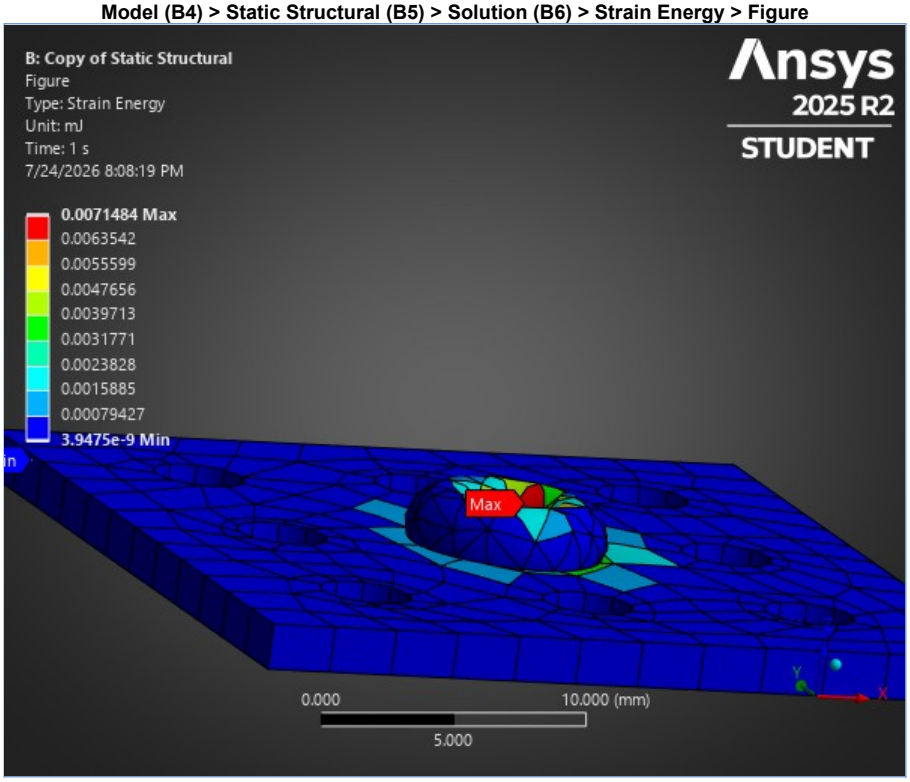


TABLE 33
Model (B4) > Static Structural (B5) > Solution (B6) > Contact Tools

Object Name	*Contact Tool*
State	Solved
Scope	
Scoping Method	Worksheet

Model (B4) > Static Structural (B5) > Solution (B6) > Contact Tool

Name	Contact Side
Frictionless - RUN 5|Body3 To RUN 5|Body11	Both

TABLE 34
Model (B4) > Static Structural (B5) > Solution (B6) > Contact Tool > Results

Object Name	*Status*
State	Solved
Definition	
Type	Status
By	Time
Display Time	Last
Calculate Time History	Yes
Identifier	
Suppressed	No
Integration Point Results	
Display Option	Averaged
Information	
Time	1. s
Load Step	1
Substep	8
Iteration Number	16

FIGURE 28
Model (B4) > Static Structural (B5) > Solution (B6) > Contact Tool > Status

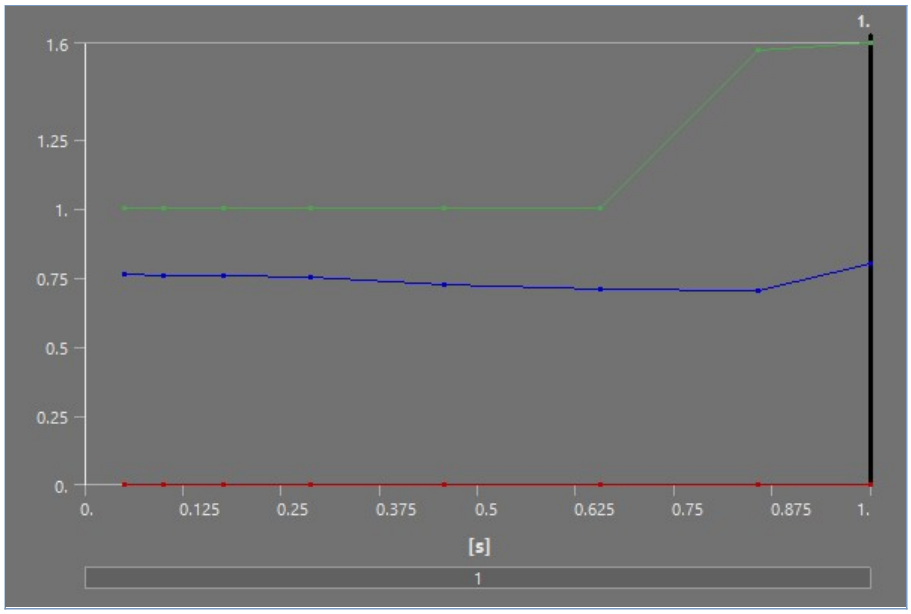


TABLE 35
Model (B4) > Static Structural (B5) > Solution (B6) > Contact Tool > Status

Time [s]	Minimum	Maximum	Average
5.e-002	0.	1.	0.75966
0.1			0.75694
0.175			0.75556
0.2875			0.75198
0.45625			0.72643
0.65625			0.70602
0.85625		1.5714	0.70424
1.		1.6	0.80254

FIGURE 29
Model (B4) > Static Structural (B5) > Solution (B6) > Contact Tool > Status > Figure

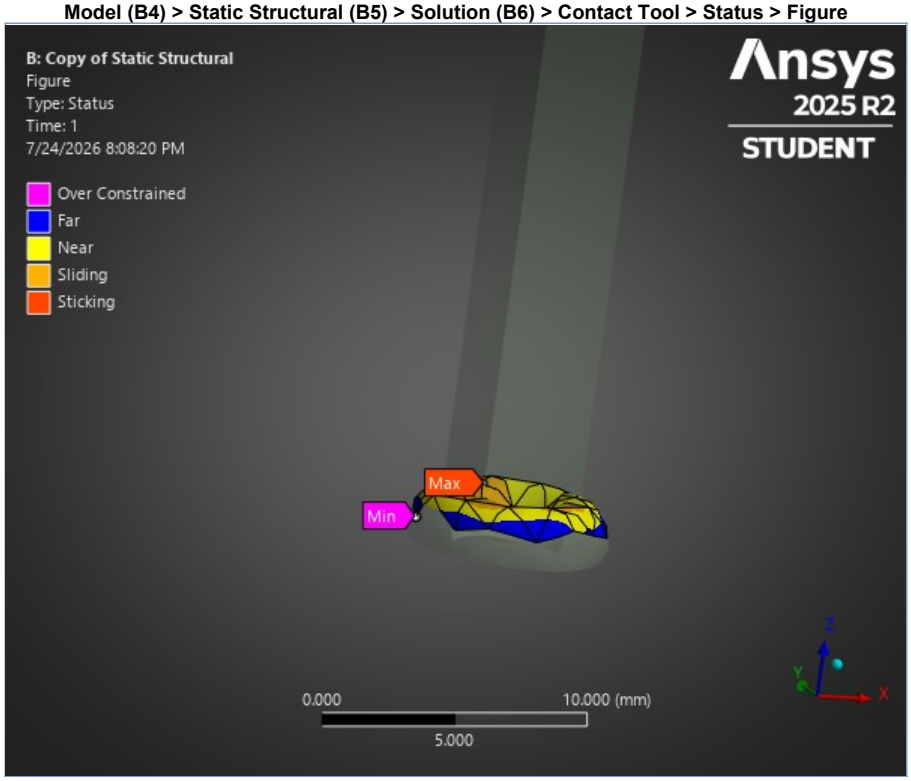


TABLE 36
Model (B4) > Static Structural (B5) > Solution (B6) > Probes

Object Name	*Force Reaction*	*Moment Reaction*
State	Solved	
Definition		
Type	Force Reaction	Moment Reaction
Location Method	Boundary Condition	
Boundary Condition	Displacement	
Orientation	Global Coordinate System	
Suppressed	No	
Summation		Centroid
Options		
Result Selection	All	
Display Time	End Time	
Results		
X Axis	9.3836e-003 N	-9.1475e-003 N·mm
Y Axis	-2.6448e-003 N	-1.0274e-003 N·mm
Z Axis	-0.52946 N	7.6518e-004 N·mm
Total	0.52955 N	9.2367e-003 N·mm
Maximum Value Over Time		
X Axis	9.3836e-003 N	-1.5271e-003 N·mm
Y Axis	-2.8748e-004 N	6.3559e-002 N·mm
Z Axis	-2.5559e-002 N	1.2346e-003 N·mm
Total	0.52955 N	6.47e-002 N·mm
Minimum Value Over Time		
X Axis	-5.9223e-004 N	-2.0828e-002 N·mm
Y Axis	-3.57e-003 N	-1.0274e-003 N·mm
Z Axis	-0.52946 N	-4.565e-004 N·mm
Total	2.5561e-002 N	4.983e-003 N·mm
Information		
Time	1. s	
Load Step	1	
Substep	8	
Iteration Number	16	

FIGURE 30
Model (B4) > Static Structural (B5) > Solution (B6) > Force Reaction

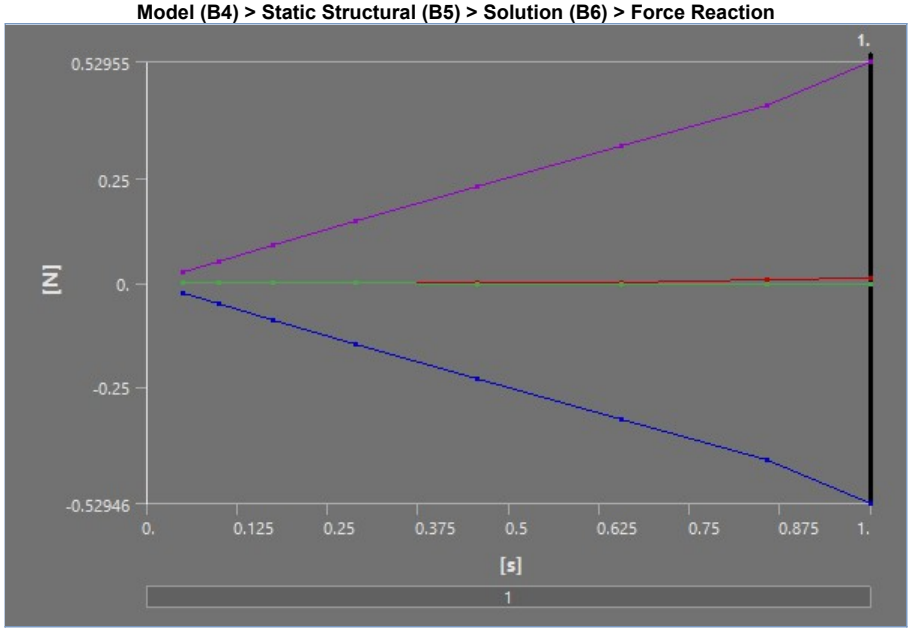


TABLE 37
Model (B4) > Static Structural (B5) > Solution (B6) > Force Reaction

Time [s]	Force Reaction (X) [N]	Force Reaction (Y) [N]	Force Reaction (Z) [N]	Force Reaction (Total) [N]
5.e-002	-1.9818e-004	-2.8748e-004	-2.5559e-002	2.5561e-002
0.1	-3.6114e-004	-5.708e-004	-5.1166e-002	5.1171e-002
0.175	-5.3171e-004	-9.8527e-004	-8.9598e-002	8.9605e-002
0.2875	-5.9223e-004	-1.5764e-003	-0.14705	0.14705
0.45625	-1.5319e-004	-2.3333e-003	-0.23158	0.23159
0.65625	1.481e-003	-2.9522e-003	-0.32635	0.32636
0.85625	5.5454e-003	-3.57e-003	-0.42595	0.426
1.	9.3836e-003	-2.6448e-003	-0.52946	0.52955

FIGURE 31
Model (B4) > Static Structural (B5) > Solution (B6) > Force Reaction > Figure

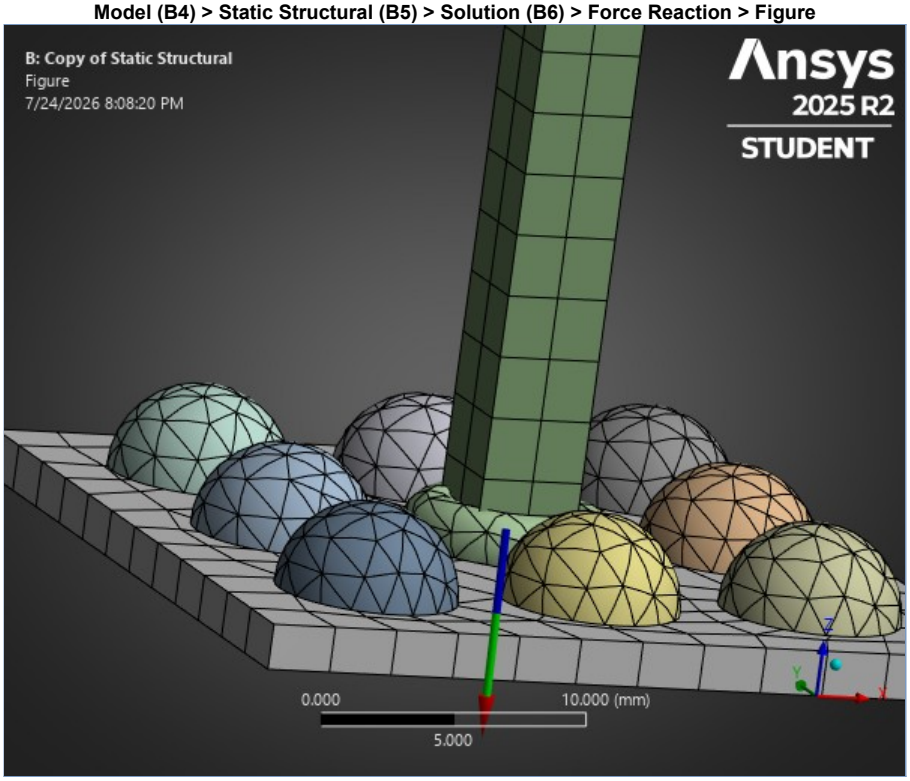


FIGURE 32
Model (B4) > Static Structural (B5) > Solution (B6) > Moment Reaction

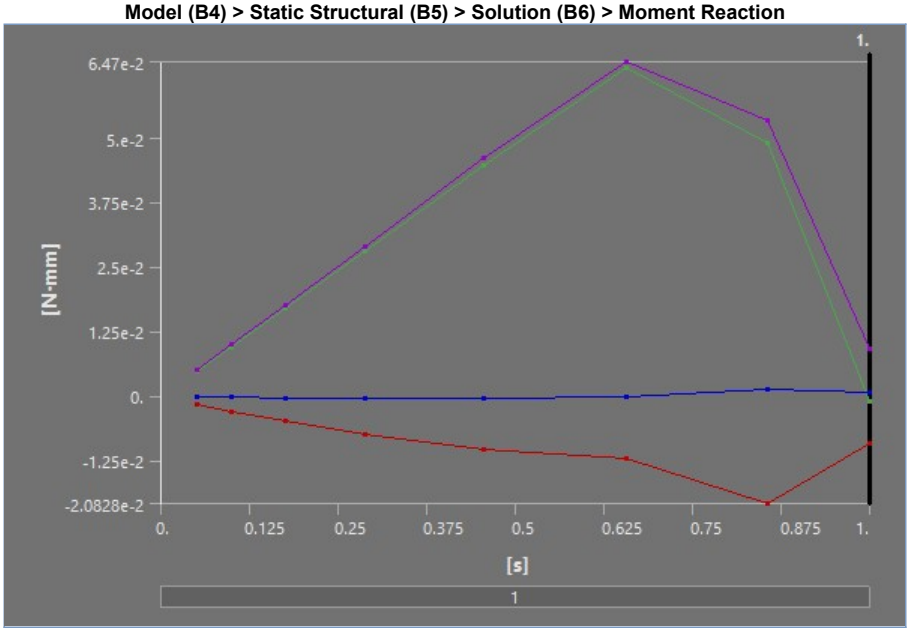
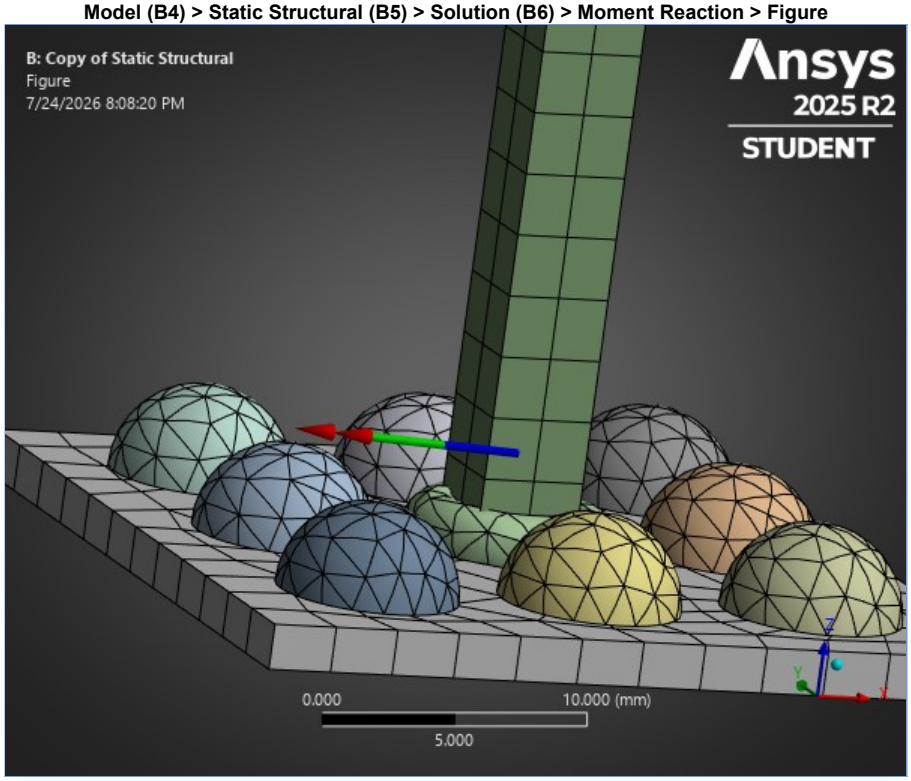


TABLE 38
Model (B4) > Static Structural (B5) > Solution (B6) > Moment Reaction

Time [s]	Moment Reaction (X) [N·mm]	Moment Reaction (Y) [N·mm]	Moment Reaction (Z) [N·mm]	Moment Reaction (Total) [N·mm]
5.e-002	-1.5271e-003	4.742e-003	-1.093e-004	4.983e-003
0.1	-2.97e-003	9.5507e-003	-2.0694e-004	1.0004e-002
0.175	-4.9679e-003	1.6871e-002	-3.2871e-004	1.759e-002
0.2875	-7.569e-003	2.8019e-002	-4.4609e-004	2.9027e-002
0.45625	-1.0464e-002	4.4769e-002	-4.565e-004	4.5978e-002
0.65625	-1.2095e-002	6.3559e-002	-1.3942e-004	6.47e-002
0.85625	-2.0828e-002	4.9034e-002	1.2346e-003	5.3289e-002
1.	-9.1475e-003	-1.0274e-003	7.6518e-004	9.2367e-003

FIGURE 33
Model (B4) > Static Structural (B5) > Solution (B6) > Moment Reaction > Figure



Material Data

ECOFLEX-0045

TABLE 39
ECOFLEX-0045 > Constants

Density	1.06e-006 kg mm^-3

TABLE 40
ECOFLEX-0045 > Color

Red	Green	Blue
182	229	228

TABLE 41
ECOFLEX-0045 > Neo-Hookean

Initial Shear Modulus Mu MPa	Incompressibility Parameter D1 MPa^-1	Temperature K
4.73e-002	0.424	

Plastic, ABS (high-impact)

TABLE 42
Plastic, ABS (high-impact) > Constants

Density	1.03e-006 kg mm^-3
Tensile Yield Strength	43.82 MPa
Tensile Ultimate Strength	36.26 MPa
Coefficient of Thermal Expansion	1.84e-004 K^-1
Thermal Conductivity	1.997e-004 W mm^-1 K^-1
Specific Heat	1.4e+006 mJ kg^-1 K^-1
Resistivity	9.95e+016 ohm mm
Electric Loss Tangent	5.916e-003
Relative Permittivity	2.993

TABLE 43
Plastic, ABS (high-impact) > Opacity

Red	Green	Blue
0	153	255

Opacity
0.9

Metallic Finish
0

TABLE 44
Plastic, ABS (high-impact) > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature K
2405	0.4089	4399.9	853.5	296.15

TABLE 45
Plastic, ABS (high-impact) > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature K
296.15

TABLE 46
Plastic, ABS (high-impact) > Climate change (CO2-eq)

Climate change (CO2-eq) dBA
3.661

TABLE 47
Plastic, ABS (high-impact) > Embodied energy

Embodied energy mJ kg^-1
9.64e+010

TABLE 48
Plastic, ABS (high-impact) > Recycle

Recycle
1